PROJECT PROPOSAL

Rafiq

Smart Tourism Companion for Egypt

DEPI – Digital Egypt Pioneers Initiative | Graduation Project 2026

1. PROJECT OVERVIEW

Project Title:

Rafiq – Smart Tourism Companion for Egypt

Objective:

To build a smart web-based platform that acts as a digital companion for tourists in Egypt, helping them plan trips, discover authentic experiences, avoid scams, and access emergency support — thereby improving Egypt's tourism image globally. The platform also empowers licensed tour guides to find job opportunities and enables tourists to communicate directly with them through an integrated marketplace and chat system.

Description:

Rafiq is a human-centered tourism platform that combines culture, safety, and technology. It provides:

- A smart trip planner tailored to budget, interests, and trip duration.
- A rich database of famous landmarks and authentic local places with cultural tips.
- A marketplace for verified tour guides with ratings, reviews, and direct booking.
- An anti-scam assistant with fair-price indicators and city-specific warnings.
- An emergency companion with one-click access to police, hospitals, and embassies.
- Smart AI features: review summarization, route optimization, and audio guides.

2. OBJECTIVES

The primary objectives of Rafiq are:

- Provide tourists with a single trusted platform covering planning, exploration, safety, and emergency support.
- Reduce tourist confusion and vulnerability to scams by offering transparent, verified information.
- Promote authentic Egyptian cultural experiences alongside well-known commercial landmarks.
- Empower licensed tour guides with a fair, transparent digital marketplace.
- Improve accessibility for tourists with disabilities through dedicated accessibility modes.
- Leverage AI and geolocation technologies to deliver smart, context-aware recommendations.

- Support Arabic and English to serve both international and local users.
- Contribute positively to Egypt's digital tourism transformation and global reputation.

3. PROJECT SCOPE

In Scope:

- Web-based platform (desktop + mobile-responsive) — no native mobile app.
- All 8 core modules: Trip Planner, Places Explorer, Guide Marketplace, Anti-Scam Assistant, Emergency Companion, Route With Soul, Smart Audio Guide, Memory Map.
- Admin dashboard to manage places, guides, and reviews.
- Bilingual interface (English & Arabic).
- Basic AI integrations: review summarization and route optimization via OpenAI API.
- Real-time guide chat using Socket.io.

Out of Scope:

- Native iOS / Android mobile applications.
- Payment gateway integration (booking is confirmed via chat — payment handled offline).
- Full offline mode.
- Third-party travel agency partnerships.

4. TEAM MEMBERS & RESPONSIBILITIES

1	**Eman**	Full-Stack Developer – Backend & Frontend Integration
2	**Basmala**	Frontend Developer – Pages, Components & Maps UI
3	**Gerges**	UI/UX Designer – Wireframes, Prototypes & Design System
4	**Mahmoud**	Frontend Developer – Marketplace, Dashboard & Admin UI
5	**Mehrael**	Backend Developer – APIs, AI Integration & Security
6	**Kirolos**	QA & Documentation – Testing, API Docs & Project Report

5. SYSTEM ARCHITECTURE & TECH STACK

Rafiq follows a client-server architecture with a React single-page application (SPA) on the frontend, a Node.js/Express REST API on the backend, and MongoDB Atlas as the primary database. Real-time communication is handled via Socket.io, and AI capabilities are provided through the OpenAI API.

Layer	Technologies
Frontend	React.js, Bootstrap 5, React Router, Redux Toolkit
Backend	Node.js, Express.js, REST API
Database	MongoDB (Atlas), Mongoose ODM
Auth	JWT, bcrypt, Role-based Access Control
AI / Smart Features	OpenAI API (review summarization, route optimization)
Maps & Geo	Google Maps API / Leaflet.js
Real-time	Socket.io (guide chat)
File Storage	Cloudinary (guide photos, place images)
Dev Tools	Git & GitHub, VS Code, Postman, Figma, MongoDB Compass
Deployment	Vercel (frontend), Render / Railway (backend)

Architecture Diagram (High Level):

- Client Layer: React SPA (Bootstrap 5, React Router, Redux) → served via Vercel CDN
- API Layer: Express.js REST API running on Node.js → deployed on Render
- Data Layer: MongoDB Atlas (cloud-hosted) + Cloudinary (media storage)
- External Services: Google Maps API, OpenAI API, Socket.io (real-time)
- Auth: JWT tokens + bcrypt password hashing + RBAC (Tourist / Guide / Admin roles)

6. METHODOLOGY

The team will follow an Agile development methodology with weekly sprints, aligned with the project timeline from February to June 2026. Each sprint will include:

- Sprint Planning – divide features into tasks and assign to team members.
- Daily async stand-ups – progress updates via WhatsApp/Discord group.
- Weekly supervisor check-ins – demo of completed features, collect feedback.
- Sprint Review & Retrospective – review deliverables and adjust the plan.

Version control is managed through Git & GitHub with a feature-branch workflow. All code is reviewed via Pull Requests before merging to the main branch.

7. PROJECT TIMELINE (FEBRUARY – JUNE 2026)

Period	Tasks & Responsibilities	Deliverables
Week 1 (Feb 1–7)	• Gerges – Create wireframes in Figma for all main pages (Home, Planner, Explorer, Marketplace) • Eman – Initialize Node.js/Express project, connect MongoDB Atlas, define folder structure • Mehrael – Research places data sources, scam patterns per city; set up backend skeleton • Mahmoud – Set up React project, Bootstrap layout, routing structure, component skeleton • Basmala – Define homepage layout and navigation flow in collaboration with Gerges • Kirls – Write project documentation plan, set up GitHub repo and branching strategy	✓ GitHub repository initialized with branching strategy ✓ Figma low-fidelity wireframes for all screens ✓ Database schema draft (Places, Users, Guides, Reviews)
Week 2 (Feb 8–14)	• Gerges – Finalize high-fidelity Figma mockups and design system (colors, typography, components) • Eman – Build Auth API (register/login/JWT), User model, role-based access control • Mehrael – Places model + CRUD API endpoints; seed initial places data • Mahmoud – Homepage + Navbar + Auth pages (Login/Register) using Bootstrap • Basmala – Places Explorer page: static UI with cards and filter layout • Kirls – Set up Postman collection, document all API endpoints planned so far	✓ Working Auth API (tested in Postman) ✓ React app running with routing and Bootstrap layout ✓ Places API (GET all, GET by ID, POST)
Week 3 (Feb 15–21)	• Gerges – Design Trip Planner UI, Places cards, cultural tips modal, and map view layout • Eman – Trip Planner full-stack: form + budget/interest filters + geo-sort algorithm + cost estimate • Mehrael – Guide model + Marketplace API (list, filter, book) + Emergency Companion data • Mahmoud – Places Explorer: dynamic cards fetched from API, search, filters, detail page	✓ Trip Planner end-to-end connected ✓ Places Explorer fetching live data ✓ Guides Marketplace API ready

	• Basmala – Trip Planner frontend: connect form to API, display results list and map pins • Kirls – QA: test Trip Planner and Places Explorer flows; log bugs on GitHub Issues	
Week 4 (Feb 22–28)	• Gerges – Design Guide Marketplace pages, booking flow, in-app chat interface • Eman – Booking API + availability calendar logic + Socket.io chat backend setup • Mehrael – Anti-Scam Assistant backend: price-warning logic, city scam data API • Mahmoud – Guide Marketplace page: listing, filter, booking form, guide profile page • Basmala – In-app chat UI + booking confirmation and success pages • Kirls – QA: test booking flow and chat; update API documentation in Postman	✓ Guide booking flow end-to-end ✓ In-app chat prototype working ✓ Anti-scam assistant API ready
Week 5 (Mar 1–7)	• Gerges – Design Audio Guide, Route With Soul, Memory Map, and Accessibility Mode UI • Eman – Smart Audio Guide full-stack: proximity trigger logic + audio content API • Mehrael – OpenAI integration: review summarizer + route optimizer + AI recommendation API • Mahmoud – Anti-Scam Assistant page + price-warning UI + negotiation phrases section • Basmala – Route With Soul: cultural route renderer on map + Memory Map shareable page • Kirls – QA: test AI features and maps; accessibility audit on all pages built so far	✓ AI-powered review summaries working ✓ Route With Soul functional on map ✓ Memory Map shareable link generated
Week 6 (Mar 8–14)	• Gerges – Design Admin Dashboard, multi-language toggle UI, and Accessibility Mode screens • Eman – Multi-language support backend logic + Arabic/English content API • Mehrael – Admin Dashboard API: manage places, guides, and reviews with moderation • Mahmoud – Admin Dashboard React pages + Accessibility Mode frontend	✓ Bilingual (AR/EN) site working ✓ Admin Dashboard operational ✓ All modules integrated and tested

	• Basmala – Multi-language toggle integration across all pages (AR/EN switch) • Kirls – Final UI polish pass, responsive QA across mobile/tablet/desktop	
Week 7 (Mar 15–21)	• Gerges – Prepare final Figma handoff file and all exported design assets • Eman – Deploy backend to Render/Railway; configure environment variables and CORS • Mehrael – Performance optimization, security hardening, rate limiting on all APIs • Mahmoud – Deploy frontend to Vercel; configure production environment • Basmala – End-to-end integration testing across all modules on production • Kirls – Write README, full API docs (Postman collection), and user guide	✓ Live deployed application (public URL) ✓ Full API documentation (Postman collection) ✓ Project README and user guide complete
Week 8 (Mar 22–31)	• Gerges – Design presentation slides and final demo screens • Eman – Fix all critical bugs found during final testing • Mehrael – Final backend stress testing and bug fixes • Mahmoud – Fix all frontend UI bugs and polish interactions • Basmala – User acceptance testing with real test users; collect feedback • Kirls – Prepare demo script, walkthrough plan, and final report draft	✓ Demo-ready application on production ✓ Graduation project presentation slides ✓ Final report first draft submitted
April – May (Supervisor Review)	• All team – Address all supervisor feedback from Week 8 review • Eman & Mehrael – Add stretch features: push notifications, advanced AI recommendations • Gerges – Update design assets based on feedback • Mahmoud & Basmala – Frontend refinements based on user testing results • Kirls – Finalize complete project documentation and defense preparation	✓ All supervisor feedback addressed ✓ Stretch features implemented (if time permits) ✓ Defense preparation materials ready
June (Final Submission)	• All team – Final code freeze and last review • Kirls – Submit final project report and all documentation	✓ Final report submitted ✓ Source code archived on GitHub

	• All team – Graduation project defense presentation	✓ Graduation project defense completed

8. EXPECTED IMPACT

- Improve Egypt's image as a modern, tourist-friendly digital destination.
- Reduce tourist fear, confusion, and vulnerability to scams.
- Promote authentic Egyptian culture beyond commercial tourist traps.
- Provide licensed tour guides with fair digital opportunities and visibility.
- Increase tourist confidence in planning and navigating Egypt independently.
- Serve as a scalable foundation for future government or private-sector tourism apps.

Rafiq

The intelligent, human-centered bridge between Egypt and its visitors — combining technology, culture, and safety in one unified platform.